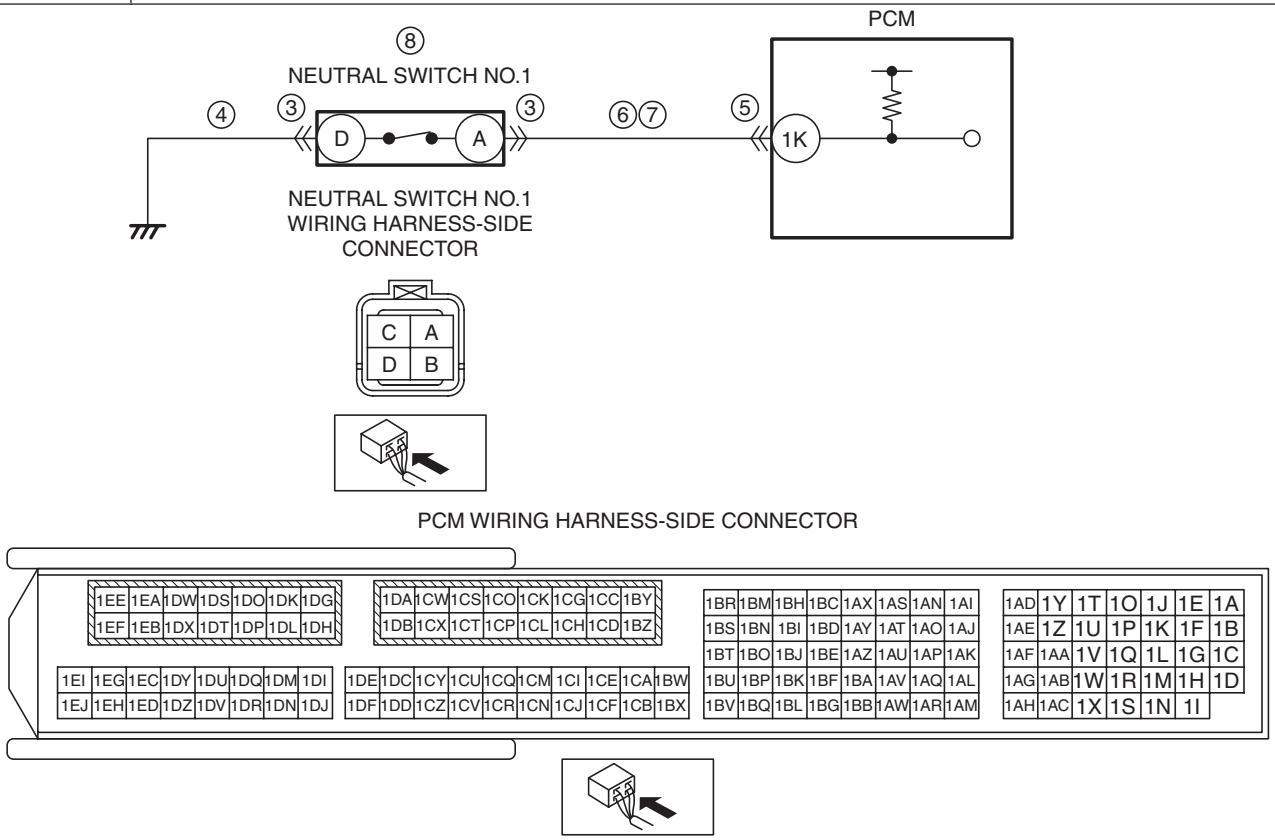
DTC P0850:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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DTC P0850:00	Neutral switch No.1 input circuit problem
DETECTION CONDITION	• The PCM monitors changes in input voltage from the neutral switch No.1. If the PCM does not detect a voltage change while driving the vehicle at a vehicle speed **above 30 km/h {19 mph}** and the clutch pedal is pressed and released **10 times** repeatedly, the PCM determines that the neutral switch No.1 circuit has a malfunction. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition in two consecutive drive cycles or in one drive cycle while the DTC for the same malfunction has been stored in the PCM. • PENDING CODE is available if the PCM detects the above malfunction condition during first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Neutral switch No.1 connector or terminals malfunction • Open circuit in wiring harness between neutral switch No.1 terminal D and body ground • PCM connector or terminals malfunction • Short to ground in wiring harness between neutral switch No.1 terminal A and PCM terminal 1K • Open circuit in wiring harness between neutral switch No.1 terminal A and PCM terminal 1K • Neutral switch No.1 malfunction • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.

STEP	INSPECTION		ACTION
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT NEUTRAL SWITCH NO.1 CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the neutral switch No.1 connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
4	**INSPECT NEUTRAL SWITCH NO.1 GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the neutral switch No.1 connector is disconnected. • Inspect for continuity between neutral switch No. 1 terminal D (wiring harness-side) and body ground. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between neutral switch No.1 terminal D and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect for the following: — Open circuit between neutral switch No.1 and body ground — Loose or lifting ground point • Repair or replace the malfunctioning part. Go to Step 9.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
6	**INSPECT NEUTRAL SWITCH NO.1 SIGNAL CIRCUIT FOR SHORT TO GROUND** • Verify that the neutral switch No.1 and PCM connectors are disconnected. • Inspect for continuity between neutral switch No. 1 terminal A (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between neutral switch No.1 terminal A and PCM terminal 1K. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.
7	**INSPECT NEUTRAL SWITCH NO.1 SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Verify that the neutral switch No.1 and PCM connectors are disconnected. • Inspect for continuity between neutral switch No. 1 terminal A (wiring harness-side) and PCM terminal 1K (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between neutral switch No.1 terminal A and PCM terminal 1K. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.

STEP	INSPECTION		ACTION
8	**INSPECT NEUTRAL SWITCH NO.1** • Inspect the neutral switch No.1. (See NEUTRAL SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Replace the neutral switch No.1, then go to the next step. (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F65M-R].) (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F66M-R].)
		No	Go to the next step.
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) **Caution** **• While performing this step, always operate the vehicle in a safe and lawful manner.** **• When the M-MDS is used to observe monitor system status while driving, be sure to have another technician with you, or record the data in the M-MDS using the PID/DATA MONITOR AND RECORD capturing function and inspect later.** • Drive the vehicle **above 30 km/h {19 mph}** and stop the vehicle. • Depress and release the clutch pedal **more than 10 times** during drive cycle. • Perform the Pending Trouble Code Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the PENDING CODE for this DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.